

Task 3 Report

Lecture Slide Generation Using LLMs

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Task:	Lecture Slide Generation Using LLMs

Objective

The objective of this task was to use Large Language Models (LLMs) to analyze an orientation lecture, understand its content, and generate a professional presentation covering the key topics discussed in the lecture.

AI Tools and Models Used

- Google AI Studio (Gemini 2.5 Pro)
- Gemini
- Claude
- PDF Conversion Tools

Complete Process Followed

Step 1: Lecture Analysis Using Google AI Studio

The orientation lecture YouTube link was provided to Google AI Studio.

Prompt 1:

Analyze this orientation lecture video. Provide: 1. Main topics discussed 2. Key concepts 3. Important announcements 4. Action items for students 5. Summary of each major section

Outcome: Google AI Studio successfully analyzed the lecture and generated a structured summary of the content. This helped in understanding the major topics and overall flow of the lecture.

Step 2: Presentation Outline Generation

After obtaining the lecture summary, a presentation outline was generated.

Prompt 2:

Using the lecture summary, create a presentation outline. Requirements: - 10 slides - Professional titles - Bullet points only - Student-friendly language

Outcome: The AI generated an initial slide outline with basic slide titles and bullet points. The outline covered the main topics but lacked presentation quality and professional structure.

Step 3: Initial Presentation Generation

A more detailed prompt was used to generate complete slide content.

Prompt 3:

Act as an expert university presentation designer. Create a professional PowerPoint presentation based on this orientation lecture. Requirements: - 12 slides - Title slide - Agenda slide - Main topics - Key takeaways - Conclusion - Speaker notes for each slide Format output slide by slide.

Outcome: The generated presentation contained the required slides and structure. However, the formatting and presentation style were not fully satisfactory.

Step 4: Evaluation of Initial Results

After reviewing the outputs generated by Google AI Studio and Gemini, several issues were identified:

- Slide formatting was inconsistent.
- Some slides contained excessive text.
- Presentation flow could be improved.
- Professional presentation quality was not fully achieved.

Therefore, further refinement was required.

Step 5: Refinement Using Claude

To improve the final output, the content generated during previous steps was collected and converted into a structured transcript. The transcript, lecture summary, and slide content were then provided to Claude for further enhancement.

Final Prompt:

Act as an expert university presentation designer. Create a professional PowerPoint presentation based on this orientation lecture. Requirements: - 12 slides - Title slide - Agenda slide - Main topics - Key takeaways - Conclusion - Speaker notes for each slide Format output slide by slide.

Outcome: Claude generated a significantly improved version of the presentation with better slide organization, improved slide titles, more concise content, professional presentation flow, enhanced readability, and better speaker notes. The final presentation was then converted into PDF format for submission.

Prompt Iterations Used

Iteration	Task	Result
Iteration 1	Lecture analysis and summarization.	Successfully extracted key information from the lecture.
Iteration 2	Presentation outline generation.	Created a basic slide structure but lacked professional quality.
Iteration 3	Professional presentation generation.	Generated complete slide content but formatting needed improvement.
Iteration 4	Content refinement using Claude.	Produced a professional and presentation-ready slide deck.

What Worked Well

- Detailed instructions
- Structured prompting
- Role-based prompting
- Step-by-step content generation
- Iterative refinement

Providing clear requirements such as slide count, presentation structure, and speaker notes significantly improved the quality of the output.

What Did Not Work Well

- Generic outputs from simple prompts
- Inconsistent slide formatting
- Excessive text on some slides
- Lack of professional presentation flow in initial versions

How Prompts Were Refined

- Added role prompting ("Act as an expert university presentation designer")
- Specified the exact number of slides
- Requested speaker notes
- Defined slide structure requirements
- Used multiple iterations instead of relying on a single prompt
- Combined outputs from multiple AI tools to achieve higher quality results

Challenges Faced and Solutions

Challenge 1: Initial Presentation Quality

Problem: Initial presentation quality was not satisfactory.

Solution: Used additional iterations and refined prompts.

Challenge 2: Slide Formatting

Problem: Slide formatting was inconsistent.

Solution: Specified formatting requirements and presentation structure.

Challenge 3: Unnecessary Details

Problem: Some outputs contained unnecessary details.

Solution: Used more focused prompts and clearer instructions.

Key Learnings

- How LLMs can analyze long-form educational content.
- The importance of prompt engineering.
- The value of iterative AI-assisted development.
- How different AI models provide different strengths.
- How combining multiple AI tools can improve final results.
- The importance of reviewing and refining AI-generated content before final submission.

Conclusion

This task provided practical experience in AI-assisted content generation and prompt engineering. By using Google AI Studio, Gemini, and Claude, I successfully analyzed an orientation lecture and transformed it into a professional presentation. The iterative prompting process significantly improved the quality of the final output and demonstrated the importance of refining prompts to achieve better results.